CO2Flux: Relative Spectral Phases of Subsurface CO_2 at Biosphere 2 LEO

A Unified Report of Documented Results and Open Validation

Dimitri Bolt
Academic Advisor: Professor Ildar R. Gabitov

28.Oct.2026

Abstract

I summarize the documented CO_2 analyses at Biosphere 2 LEO without changing my approved research plan. For vertical W_x1_y4, the accepted observation interval is 01.May.2025–30.Sep.2026 (17 complete months). The limited Stage 0.1-B hydrological input consists of 14 series and 37 indirect wetting dates, not a verified irrigation-event calendar. Stage 0.2 is closed using my monthly median daily-amplitude threshold of 2.5 in each channel's stored units: 6,123 common two-hour cells and 489 technically computable 30-day windows. My older four-channel phase demonstration comprises 492 windows obtained with previous input masks; I have not recomputed those four-channel phases on the accepted Stage 0.2 input. A newer Stage 0.5 diagnostic compares 15-minute and two-hour subsurface Fourier phases across 489 matched windows for two adjacent depth pairs. It separates the additional common-cell restriction caused by the atmospheric sensor. I report both numerical sensitivity and limitations without deciding whether two-hour sampling is adequate. The final Stage 0.5 decision by I. R. Gabitov and the conditions for new Chapter 1 remain open.

Contents

Research roadmap and source correspondence 3

1 Introduction and research status 4

2 LEO experimental setup 4

3 Hydrological context: accepted Stage 0.1-B input 4

4 Data and quality control 4
4.1 Accepted vertical and scientific scope of Stage 0.2 . . . 4
4.2 Common cells and preserved missingness . . . 6
4.3 Older pilot results remain distinguished . . . 6

5 Fourier methodology 6
5.1 Stage 0.3: amplitude spectra . . . 6
5.2 Stage 0.4: direct complex coefficients and relative phases . . . 8
5.3 Computational implementation and validation . . . 8

6 Preliminary results 8
6.1 Three temporal series . . . 8
6.2 One 5-to-20 cm discontinuity examined for Stage 0.5/Q1 . . . 10
6.3 Stage 0.5: sensitivity to 15-minute versus two-hour grids . . . 10

7 Historical analyses, not spectral-phase evidence **13**

8 Mathematical validation and physical interpretation **13**

9 Buchstaber framework: approved methodological preparation only **15**

10 Limitations and open questions **15**

11 Next approved steps **16**

Sources, figures and reproducibility **16**

Research roadmap and source correspondence

The identifiers below follow the existing research README and the approved PLAN. They identify research stages, not the numbered sections of this report. In particular, **0.5/Q1** is a completed diagnostic of one phase change, *not* completion of Stage 0.5. Existing numbered historical notebooks concern the previous methodology, not the future spectral chapters.

Stage	Object and status	Report section	Canonical source
0.1-B	Limited hydrological input accepted; full event log missing	Hydrology	Hydrological audit
0.2	Accepted 17-month input and author threshold 2.5; closed with limitations	Data and QC	Data checks
0.3	Preliminary amplitude spectra; calculated, not a final frequency choice	Amplitude spectra	Spectra notebook
0.4	Earlier 492-window phase demonstration, not recalculated with 489-window input	Direct Fourier; phase series	Phase notebook, sections 1–9
0.5/Q1	One phase jump analyzed; does not close Stage 0.5	Single-jump result	Phase notebook, section 10
0.5/grid	15-minute versus two-hour phases for two subsurface pairs; diagnostic completed	Grid comparison	Phase notebook, section 11
0.5	Presentation to I.G. and explicit approval; pending	Next approved steps	PLAN: 0.5 and stopping criteria
1–3	Future spectral-phase chapters; not yet begun	Next approved steps	PLAN: later stages
4	Buchstaber method; methodological preparation only	Buchstaber	Buchstaber notebook
5–6	Spatial links and physical mechanisms; future work	Next approved steps	PLAN: later stages
Historic 1–3	Absolute-harmonic maximum times/amplitudes; distinct earlier work	Historical analyses	Notebook 01, 02, 03

1 Introduction and research status

The long-term objective is to assess whether changes in CO_2 accompanying the transformation of crushed basalt into soil allow transport, chemical interactions and biological sources or sinks to be distinguished. The immediate approved observable is instead precisely defined: three relative spectral phases between consecutive levels. Absolute times of fitted diurnal maxima are not interchangeable with these quantities. The current canonical README and PLAN place the project at preparatory Stage 0, before the new Chapter 1 [1, 2]. This report assembles existing results, not new scientific computations or an alternative research plan.

2 LEO experimental setup

The selected pilot vertical on LEO West is at $(x, y) = (1, 4)$ m. Four channels are used: air, 5, 20 and 50 cm. The air LI-COR instrument is at $(0, 4)$ m, horizontally 1 m from the basalt profile. Its nominal height +0.25 m comes from Excel metadata and is not independently established by the Oracle BOXZ field. The assumption $C_0(t) = C_{\text{air}}(t)$ is adopted for the present analysis and must not be mistaken for experimental proof of an exact surface boundary condition [4, 5].

Level	Sensor	Model	Stored unit
Air	1275	LI-COR	μmol/mol
5 cm	996	GMM222	ppm
20 cm	1012	GMM222	ppm
50 cm	1035	GMM222	ppm

The historical West-slope profile W_R4_C-4 is different: sensors 994/1010/1026 at 5/20/35 cm. Its results are not pooled with this four-channel 5/20/50 cm pilot.

3 Hydrological context: accepted Stage 0.1-B input

For the interval 01.May.2025–30.Sep.2026 I accepted 14 hydrological time series: two 5TM volumetric-water-content (VWC) channels, two MPS-2 matric-potential channels, and ten separate Model3130 stored-mass channels. The existing source export contains 696,136 rows, including 289 service values of −9999. The original time stamps, sensor identifiers, real gaps, service values, and monthly coverage are preserved; no missing concentrations or hydrological measurements were interpolated [9, 10].

VWC sensors 19/317 (%) are at neighboring coordinate $(2, 4)$, depths 5 and 35 cm, approximately one meter from my CO_2 vertical. MPS-2 sensors 668/889 (kPa) are at W_x1_y4, depths 20 and 50 cm. The ten stored masses 1066–1075 (kg) refer to support locations across the slope, and I do not equate them with the amount of supplied water or the water content of the selected CO_2 profile. No conversion from matric potential to VWC is made without an appropriate water-retention relationship.

I retained **37 calendar dates of independently corroborated wetting**. These are indirect observations, not 37 verified operational irrigation events. A complete SCADA log specifying start/end times, zone, water volume and flow has not been recovered. Thus Stage 0.1-B is complete for the authorized pilot, while the original full-event-calendar question remains open.

4 Data and quality control

4.1 Accepted vertical and scientific scope of Stage 0.2

I selected W_x1_y4 with LI-COR 1275 (air) and GMM222 996/1012/1035 (5, 20, 50 cm). After checking all 16 candidate West-slope verticals, I closed Stage 0.2 on 23.Oct.2026 with the half-open interval

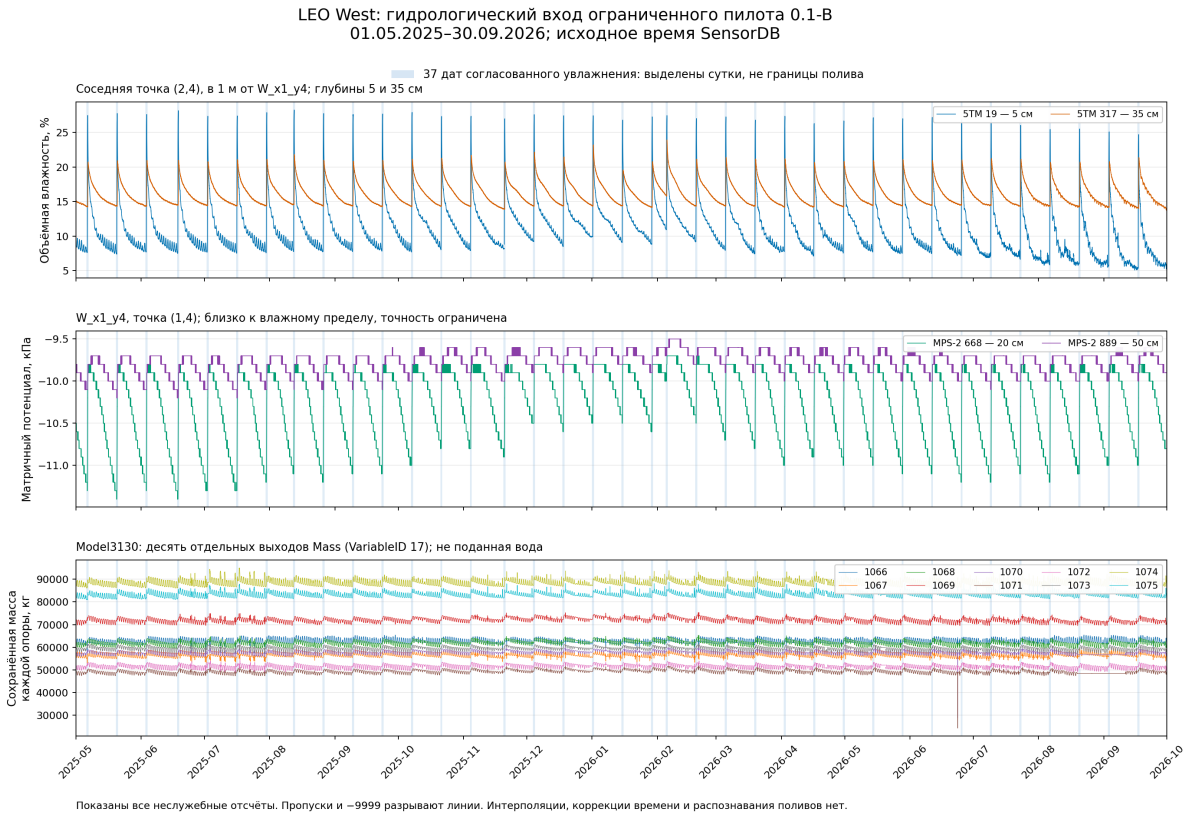


Figure 1: Existing 17-month hydrological overview: neighboring VWC, on-vertical matric potential and ten separate slope-support masses. The 37 marked days identify wetting evidence, not exact irrigation events.

[01.May.2025, 01.Oct.2026): **17 consecutive complete months** with actual data gaps retained [11, 12].

I approved a minimum monthly median diurnal amplitude of **2.5** in each channel's original units: ppm for GMM222 and μmol/mol for LI-COR. This was my separate decision and does *not* replace or claim to be I. R. Gabitov's original 8-unit threshold. The strict-passport result at 8 remains unchanged. My accepted Fourier input selector is `technically_retained & ~doubtful`; it preserves the original observations and rejection reasons elsewhere. The accepted database time reference is not shifted.

On this view I obtained **517 of 518 common days** and **6,123 of 6,216** possible two-hour common cells (98.50%). The removal of doubtful values lost 25 previously occupied common cells. The accepted 30-day / one-day-step calendar has **489 technically computable windows**, each with 316–360 common two-hour cells. This is a data-availability result: it does not mean 489 independently corroborated four-channel phase calculations. I have not introduced new R^2, coverage, or Fourier-modulus thresholds.

4.2 Common cells and preserved missingness

For a channel j and calendar cell $B_k = [s_k, s_k + 2\,\mathrm{h})$, let R_{jk} be its actually accepted source measurements. I define

$$n_{jk} = |R_{jk}|, \qquad x_{jk} = \frac{1}{n_{jk}} \sum_{r \in R_{jk}} C_{jr} \quad (n_{jk} > 0). \tag{1}$$

For window W I use only shared occupied cells

$$I_W = \{k : B_k \subset W,\ n_{jk} > 0 \text{ for all } j = 0, 1, 2, 3\}, \qquad N = |I_W|. \tag{2}$$

No absent cell is replaced by zero, no interpolation or unlimited atmospheric forward-fill is performed, and the actual calendar is not compressed. The 15-minute comparison below uses separate common cells of the three subsurface channels.

4.3 Older pilot results remain distinguished

The surviving Stage 0.3 spectra and original Stage 0.4 figures use an earlier calendar beginning 29.Apr.2025 and older technical masks. That pilot had 6,184 occupied common two-hour cells out of 6,252 possible and computed 492 windows. The accepted Stage 0.2 view instead has 6,123/6,216 cells and 489 potential windows. I retain those historical preliminary plots *without claiming a recalculation of four-channel Fourier phases or amplitude spectra using the newly accepted Stage 0.2 masks* [3, 5, 6, 11].

5 Fourier methodology

5.1 Stage 0.3: amplitude spectra

My previously saved preliminary spectra use a floating-intercept, equal-weight generalized Lomb–Scargle sinusoidal least-squares model at each candidate frequency:

$$C_j(t) \simeq c_j(f) + a_j(f)\cos(2\pi f t) + b_j(f)\sin(2\pi f t) = c_j(f) + \mathrm{Re}\{Z_j(f)e^{2\pi i f t}\}, \quad Z_j = a_j - ib_j, \quad A_j(f) = |Z_j|. \tag{3}$$

A_j is a fitted sinusoidal semi-amplitude in the source unit, not spectral power. This is not the direct Fourier estimator employed in Stage 0.4. The existing method is implemented by `gls(t,y,frequency)` and `grid(days)` in `Project_description/sensorDB/preliminary_spectra.py`. The frequency step is $1/T \simeq 0.001921153$ cycles/day for an observed span $T \simeq 520.5208$ days, over $0 < f < 6$ cycles/day. The display bound is not a strict Nyquist claim for irregular sampling. Other spectral peaks are documented, but their mechanisms and an optimal physical frequency have not been established. The 24-hour harmonic remains my provisional demonstration frequency [3, 8].

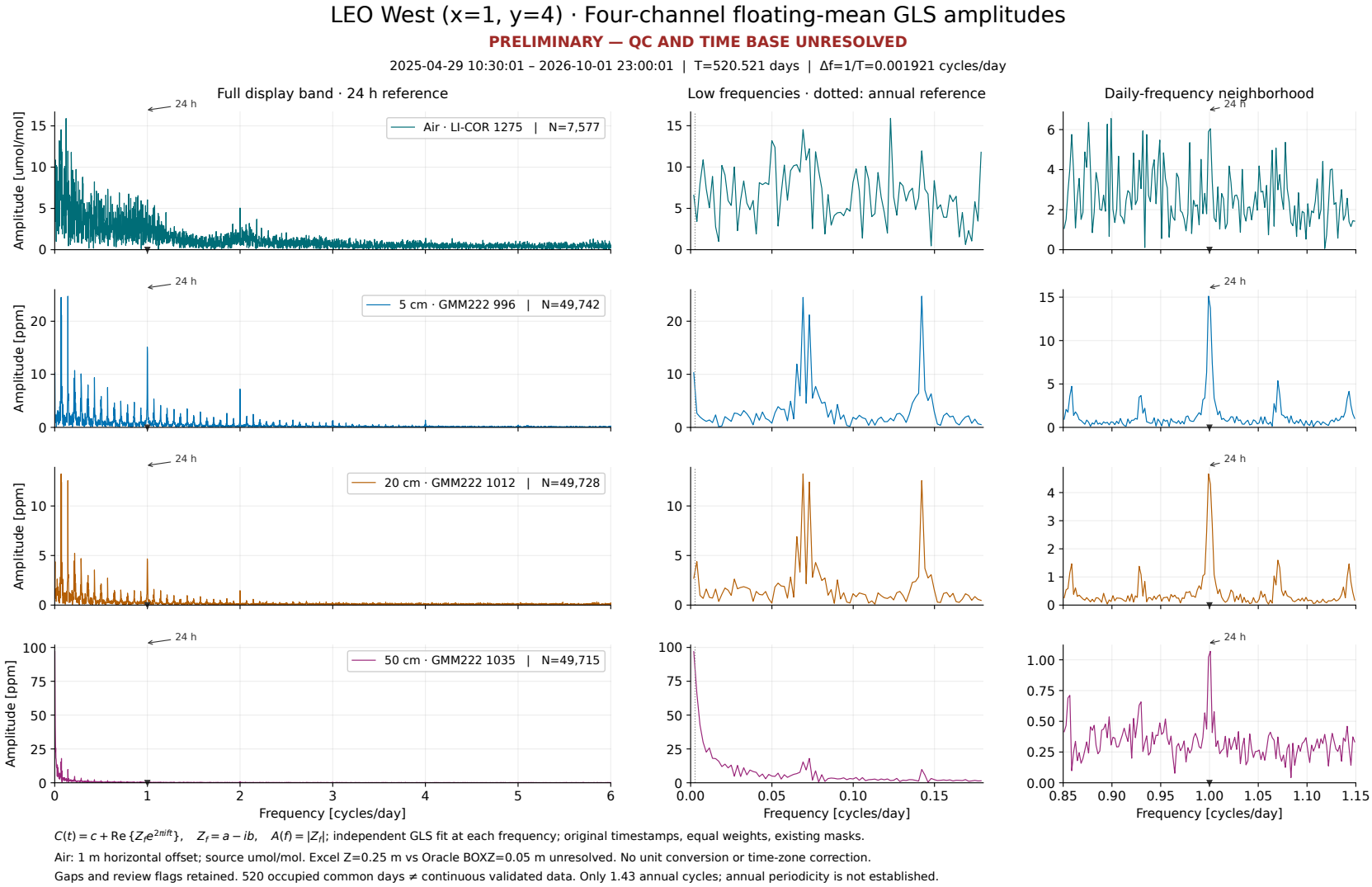


Figure 2: Pre-existing four-channel preliminary amplitude spectra, obtained by independent floating-mean sinusoidal fits. A daily-frequency marker does not constitute frequency selection or phase validation.

5.2 Stage 0.4: direct complex coefficients and relative phases

For the older Stage 0.4 demonstration I set t_{ref} =29.Apr.2025 00:00. For each common two-hour center $t_k = s_k + 1$ hour, express $t_k - t_{\mathrm{ref}}$ in days. The working frequency is $f_0 = 1$ cycle/day, window length 30 days, and step one day. For channel j use

$$\bar{x}_{j,W} = \frac{1}{N} \sum_{k \in I_W} x_{jk}, \tag{4}$$

$$F_j(f_0; W) = \frac{1}{N} \sum_{k \in I_W} (x_{jk} - \bar{x}_{j,W})\, e^{-2\pi i f_0 t_k}. \tag{5}$$

Here t_k is understood relative to the common time origin. Each occupied two-hour cell receives equal weight regardless of its raw observation count. No additional detrending or taper is applied. This is a direct complex Fourier sum on the actual common timestamps, not a gap-compressed FFT or a GLS fit. For ordered neighboring levels $u \to v$ define

$$Q_{uv} = F_u \overline{F_v}, \qquad \Delta\phi_{u\to v} = \mathrm{wrap}(\mathrm{Arg}\, Q_{uv}), \tag{6}$$

$$\mathrm{wrap}(\alpha) = (\alpha + \pi) \bmod (2\pi) - \pi \in [-\pi, \pi). \tag{7}$$

Thus $\Delta\phi_{\mathrm{air}\to 5}$, $\Delta\phi_{5\to 20}$ and $\Delta\phi_{20\to 50}$ are the report's three principal observables. For ideal sinusoidal pairs $u(t) = \cos(\omega_0 t)$ and $v(t) = \cos(\omega_0(t - \tau))$ on a complete balanced grid, where $\omega_0 = 2\pi f_0$, one has $F_u \propto 1$ and $F_v \propto e^{-i\omega_0\tau}$; hence

$$\Delta\phi_{u\to v} = \mathrm{wrap}(\omega_0 \tau). \tag{8}$$

A positive phase denotes a lag of the second *ideal sinusoid*, modulo 24 hours. This identity does not establish a gas-transport lag or transport direction in LEO.

5.3 Computational implementation and validation

The existing `Project_description/sensorDB/preliminary_fourier_phases.py` defines `aggregate_bins(frames,` `direct_fourier(t_days,x,f0=F0)`, `phase_difference(f_upper,f_lower)`, `wrap(phi)` and `compute_windows(gr` The actual implementation uses `pandas.Series.dt.floor`, `DataFrame.groupby(...).agg`, `reindex`, and `numpy.mean`, `numpy.exp`, `numpy.sum`, `numpy.conj`, `numpy.angle`. This is a verifiable mapping to existing Python methods, not a new listing.

The saved synthetic checks recover zero for identical signals, $+\pi/3$ for a four-hour delay of the second sinusoid, $-\pi/3$ when channels are reversed and $-5\pi/6$ for a 14-hour delay wrapped from $7\pi/6$. The selected irregular common-bin mask gives 1.0515280496203037 rad rather than $\pi/3$, an error of 0.004330498423706031 rad in that synthetic case. On a complete synthetic grid, the maximum complex difference from a correctly phase-corrected `numpy.fft.fft` result was 1.9421×10^{-15}. The original validation recorded 13 passing tests; the later Q1 diagnostic recorded 19 passing tests in total. These test results verify arithmetic and sign conventions, not physical sensor reliability [4, 6, 7].

6 Preliminary results

6.1 Three temporal series

The older Stage 0.4 demonstration computed three relative phases in 492 overlapping 30-day windows, with centers 14.May.2025–17.Sep.2026. These results retain the earlier masks and are not a 489-window reanalysis on the new accepted Stage 0.2 input. The coefficient moduli, missingness and approval of phase-admission criteria must be considered before physical interpretation.

The three angular coordinates define a candidate point in $\mathbb{T}^3 = S^1 \times S^1 \times S^1$. The cube $[-\pi, \pi)^3$ is only one coordinate display with opposite faces identified; a plotted branch jump or Euclidean cube distance is not automatically intrinsic motion on the torus.

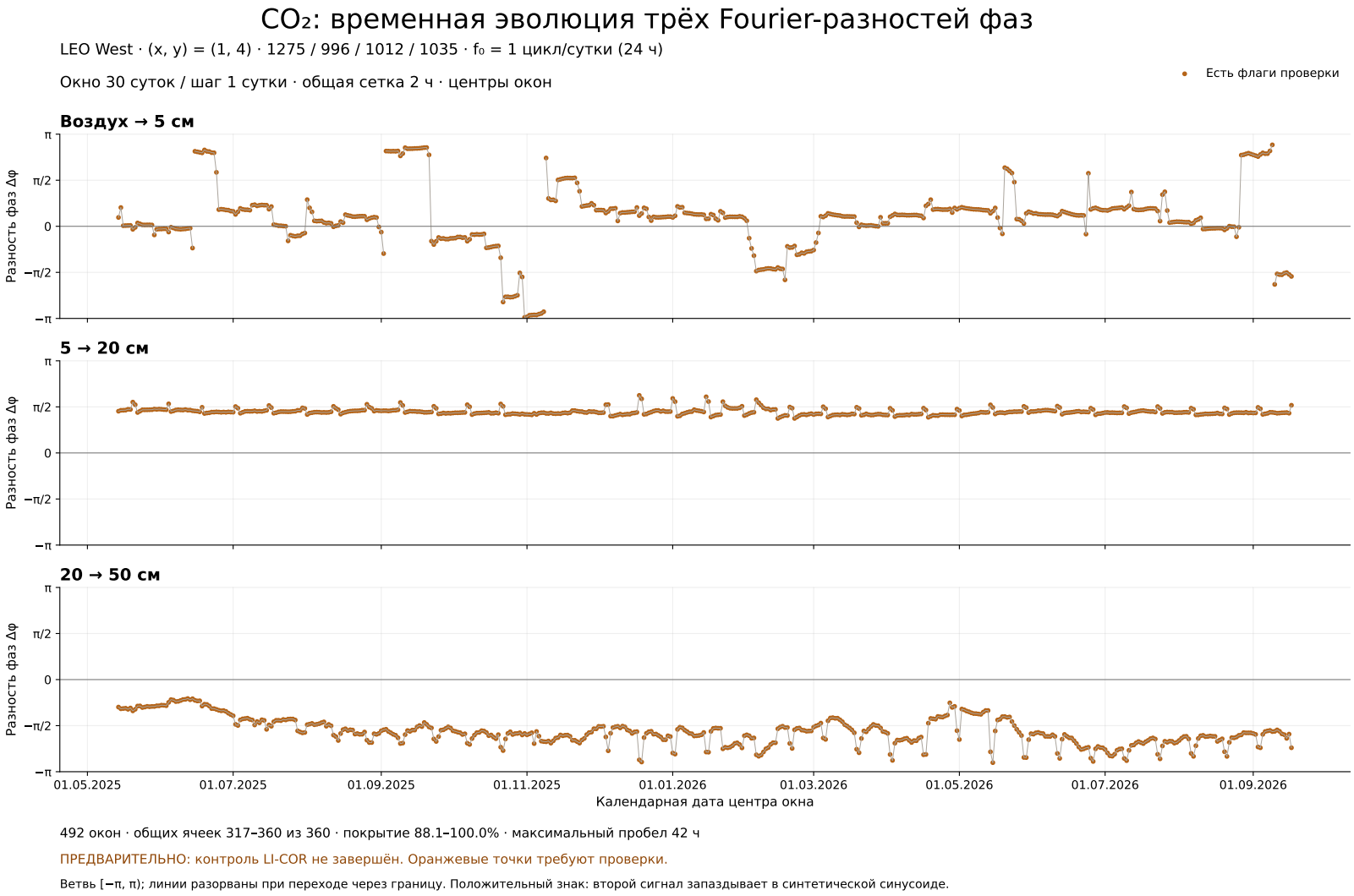


Figure 3: Pre-existing time evolution of three direct-Fourier relative phases, 30-day windows, one-day step, one cycle/day. Computed on the earlier input masks; not updated to the Stage 0.2 selected input.

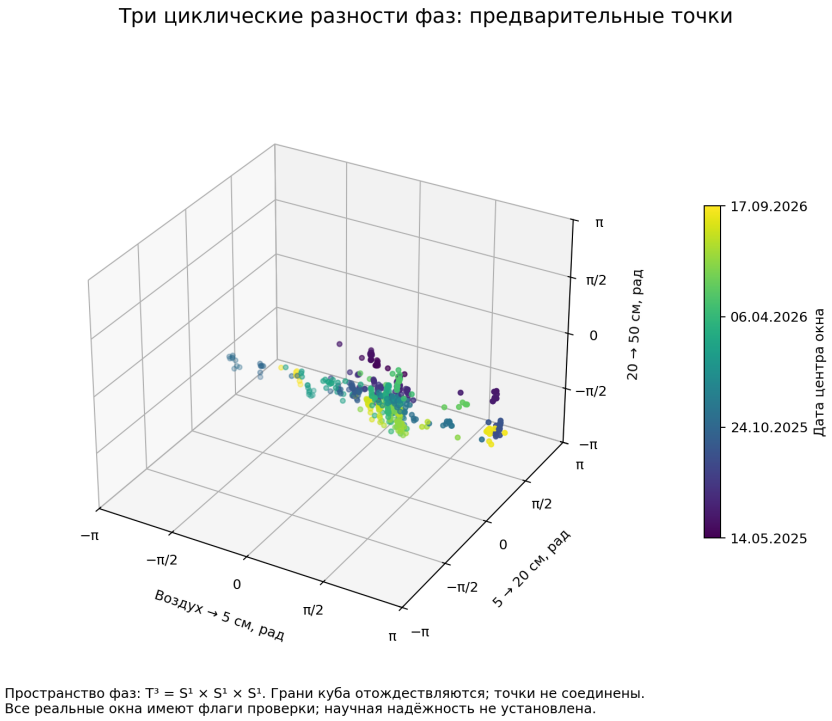


Figure 4: Pre-existing visualization of the three cyclic phase coordinates; it is not a demonstrated physical trajectory or a validated topological invariant.

6.2 One 5-to-20 cm discontinuity examined for Stage 0.5/Q1

My earlier 0.5/Q1 diagnostic selected a single adjacent pair by maximal absolute difference on the original $[-\pi, \pi)$ phase branch, with earlier occurrence breaking ties. The centers are 17.Dec.2025 and 18.Dec.2025, corresponding to 30-day windows from 02.Dec.2025 through 01.Jan.2026 and from 03.Dec.2025 through 02.Jan.2026, respectively. Neighboring windows overlap for 29 days and must not be treated as independent events. For the complex product $Q = F_5\overline{F_{20}}$, distinguish

$$d_Q = \mathrm{wrap}(\mathrm{Arg}\, Q_2) - \mathrm{wrap}(\mathrm{Arg}\, Q_1), \qquad \delta_Q = \mathrm{wrap}(\mathrm{Arg}(Q_2\overline{Q_1})), \qquad d_Q - \delta_Q = 2\pi k,\ k \in \mathbb{Z}. \tag{9}$$

The saved phases are 1.3815969902 and 1.9625409175 rad, so $d_Q = \delta_Q = +0.5809439273$ rad and $k = 0$: *this change is not an argument-branch crossing.* The coefficient argument changes are +0.0891610601 rad for F_5 and −0.4917828672 rad for F_{20}, agreeing with the product-phase increment. Their moduli decrease from 10.087605 to 6.749558 ppm and from 2.715018 to 2.004241 ppm. All eight neighboring windows have 360/360 common bins, but all contain review flags. None of these sampled coefficients is zero. There is no established amplitude-error threshold and no knowledge of an intervening continuous coefficient path, so a zero crossing cannot be proved or excluded. No physical event or mechanism is established [4, 7].

6.3 Stage 0.5: sensitivity to 15-minute versus two-hour grids

After discussion with I. R. Gabitov on 23.Oct.2026, I undertook one restricted diagnostic of the *subsurface* relative phases 5→20 cm and 20→50 cm. I used the accepted 17-month Stage 0.2 source observations, $f_0 = 1$ cycle/day, and matching 30-day windows stepped by one day. I distinguished three supports: (i)

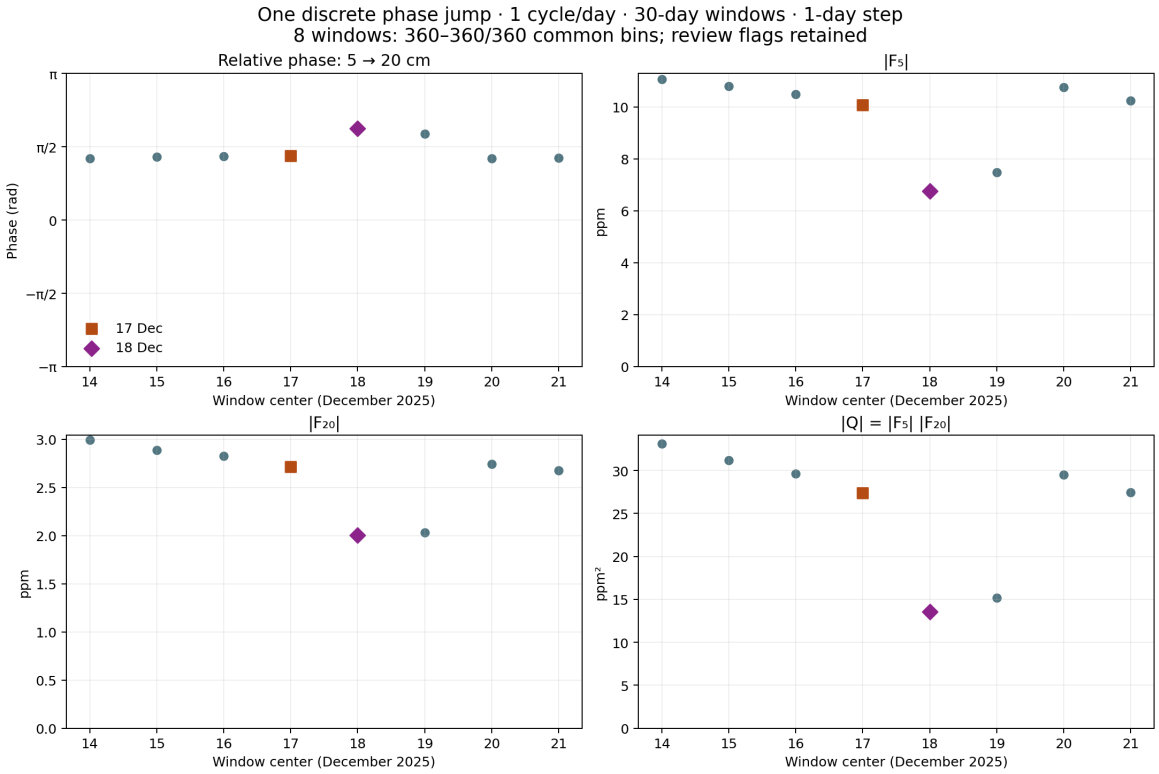


Figure 5: Saved discrete local relative phase and coefficient moduli around one selected phase change.

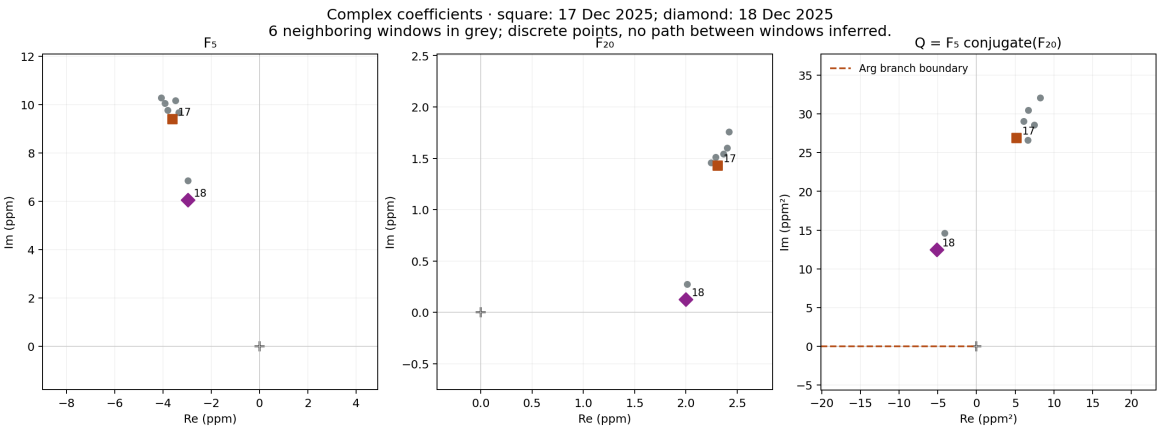


Figure 6: Saved complex Fourier coefficients and their product Q, showing the origin and the relevant argument branch boundary without interpolating an unknown path.

common 15-minute subsurface bins; (ii) common two-hour subsurface bins; (iii) two-hour subsurface means further restricted to cells containing an actual atmospheric LI-COR reading. This last operation is *not* the same as aggregation, and does not reconstruct a 15-minute air signal [13, 14].

For the aggregation comparison I calculate the cyclic difference

$$\delta_{u\to v}(W) = \operatorname{Arg}\left[\exp\left(i\left\{\Delta\phi_{u\to v}^{(15\,\mathrm{min})}(W) - \Delta\phi_{u\to v}^{(2\,\mathrm{h})}(W)\right\}\right)\right]. \tag{10}$$

I calculate an analogous circular difference between the unrestricted two-hour subsurface phase and the phase on two-hour cells also occupied by air. All **489 windows** yielded finite paired estimates for both subsurface depth pairs in all three variants.

Effect	Pair	Median $\lvert\delta\rvert$	95th pct. $\lvert\delta\rvert$	Maximum $\lvert\delta\rvert$
15 min – 2 h	5→20 cm	0.010179	0.050569	0.080158
15 min – 2 h	20→50 cm	0.050098	0.154327	0.317131
Air support	5→20 cm	0.011741	0.315633	0.430516
Air support	20→50 cm	0.052175	0.297058	0.399709

All values are in radians. The 15-minute subsurface windows have 2,776–2,831 occupied common cells out of 2,880 possible; two-hour subsurface support has 360/360, and the two-hour support additionally restricted by air has 316–360/360. The respective longest missing common-cell intervals are 1, 0 and 42 hours. The preserved validation records **26 passing tests**, including seven new narrow checks; it does not establish physical uncertainty or a new acceptance threshold.

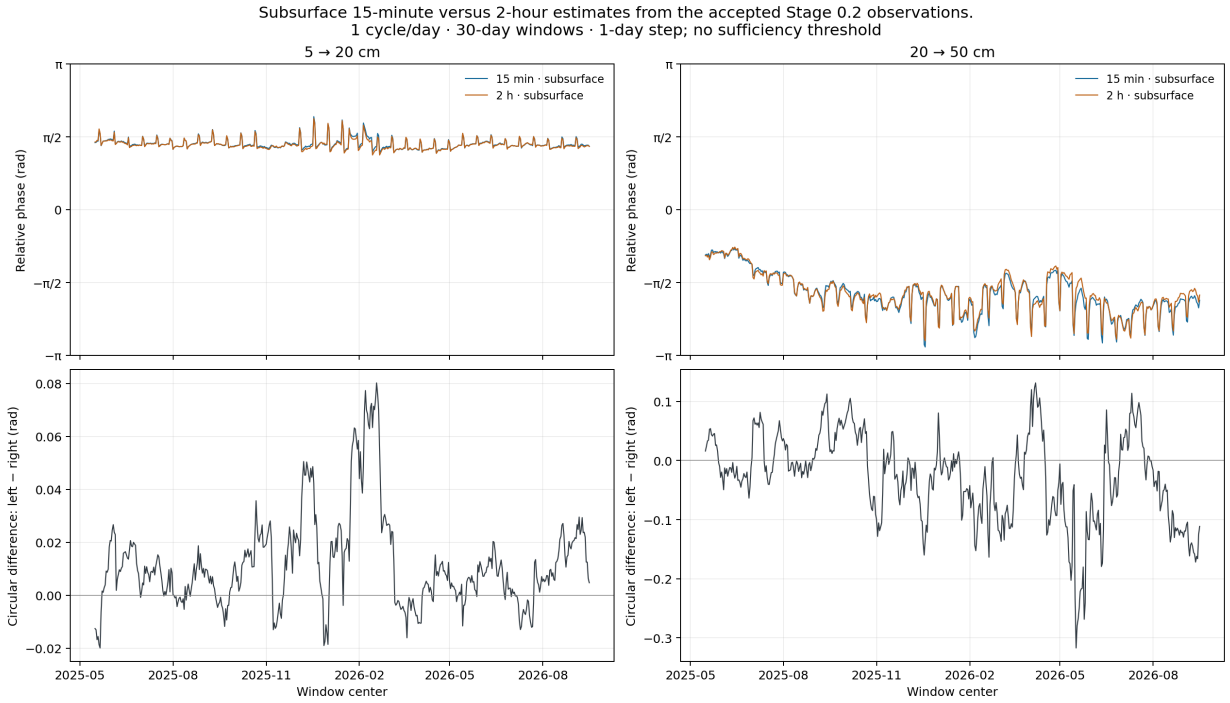


Figure 7: Existing comparison between actual subsurface 15-minute and two-hour Fourier phases on the same accepted measurements. The lower panels show circular differences between estimates, not errors relative to an unknown true phase.

I interpret this calculation as a check of sampling and common-support sensitivity for two subsurface pairs at one daily frequency. Overlapping windows are not independent observations. The sufficiency of two-hour resolution remains a matter for an explicit methodological decision by I. R. Gabitov.

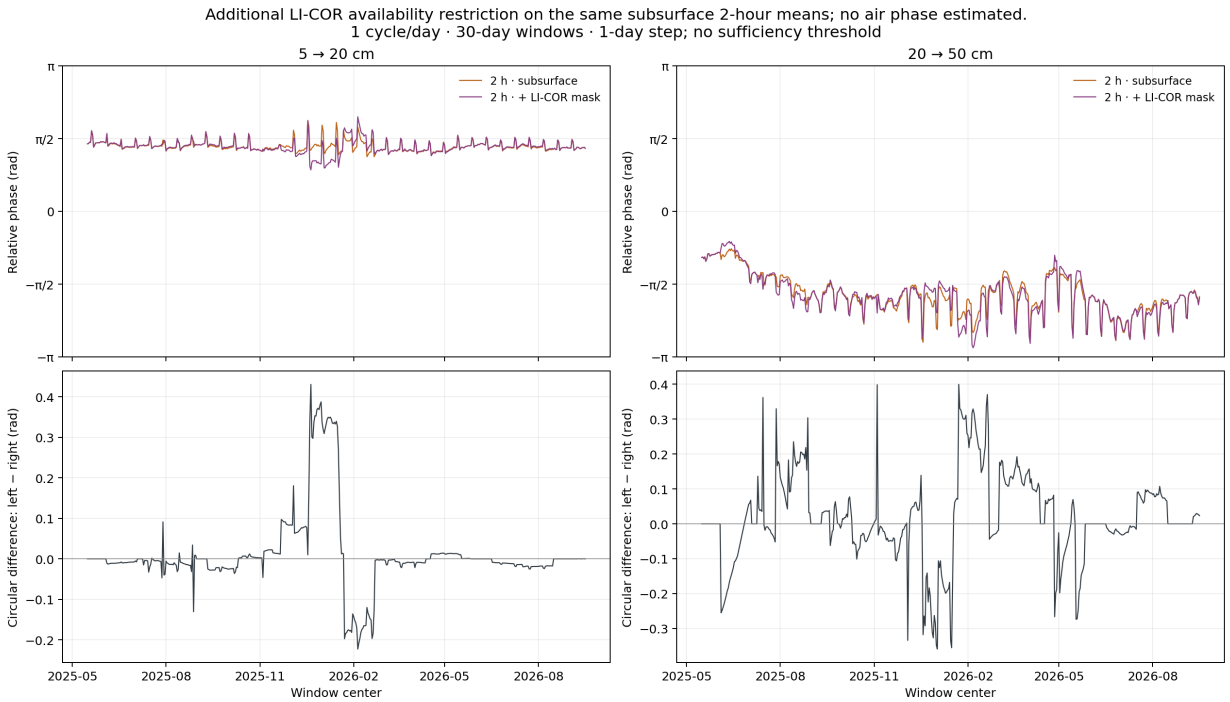


Figure 8: Existing effect of restricting two-hour subsurface support by atmospheric LI-COR availability. The two-hour subsurface means themselves are preserved; this is not a 15-minute atmospheric phase analysis.

7 Historical analyses, not spectral-phase evidence

In my earlier Chapter 1, I fitted a fixed diurnal harmonic. My earlier Chapter 2 studied the different West vertical W_R4_C-4 from 11.Sep.2013 through 30.Sep.2026, over 157 months and 2,614,638 archived raw measurements. Its final historical processing retained 43 jointly admitted months, 23 vectors of absolute daily maximum times $H(d)$ and 25 daily-amplitude vectors $A(d)$. These are not four-channel spectral-window admissions. My earlier Chapter 3 found no established reproducible long-term evolution under its original absolute-maximum-time and amplitude procedure. The historical input and its limitations are retained but are not recalculated or interpreted as either confirming or refuting the new relative spectral-phase problem [1, 15].

8 Mathematical validation and physical interpretation

I treat the argument of a nonzero complex Fourier coefficient as a computed angular coordinate, not automatically as a reliable physical travel time. In particular, small moduli can make phase sensitive to perturbation, and changing the common observation support can change the result. The single older Q1 jump and the new grid-resolution diagnostic test different mathematical issues. Neither establishes a CO_2 transport mechanism.

For subsequent physical interpretation, the PLAN records an elementary one-dimensional diffusion hypothesis discussed with I. R. Gabitov on 23.Oct.2026:

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}, \qquad D > 0, \tag{11}$$

where x is depth directed downward. This has not been fitted or validated for LEO. A possible, not yet

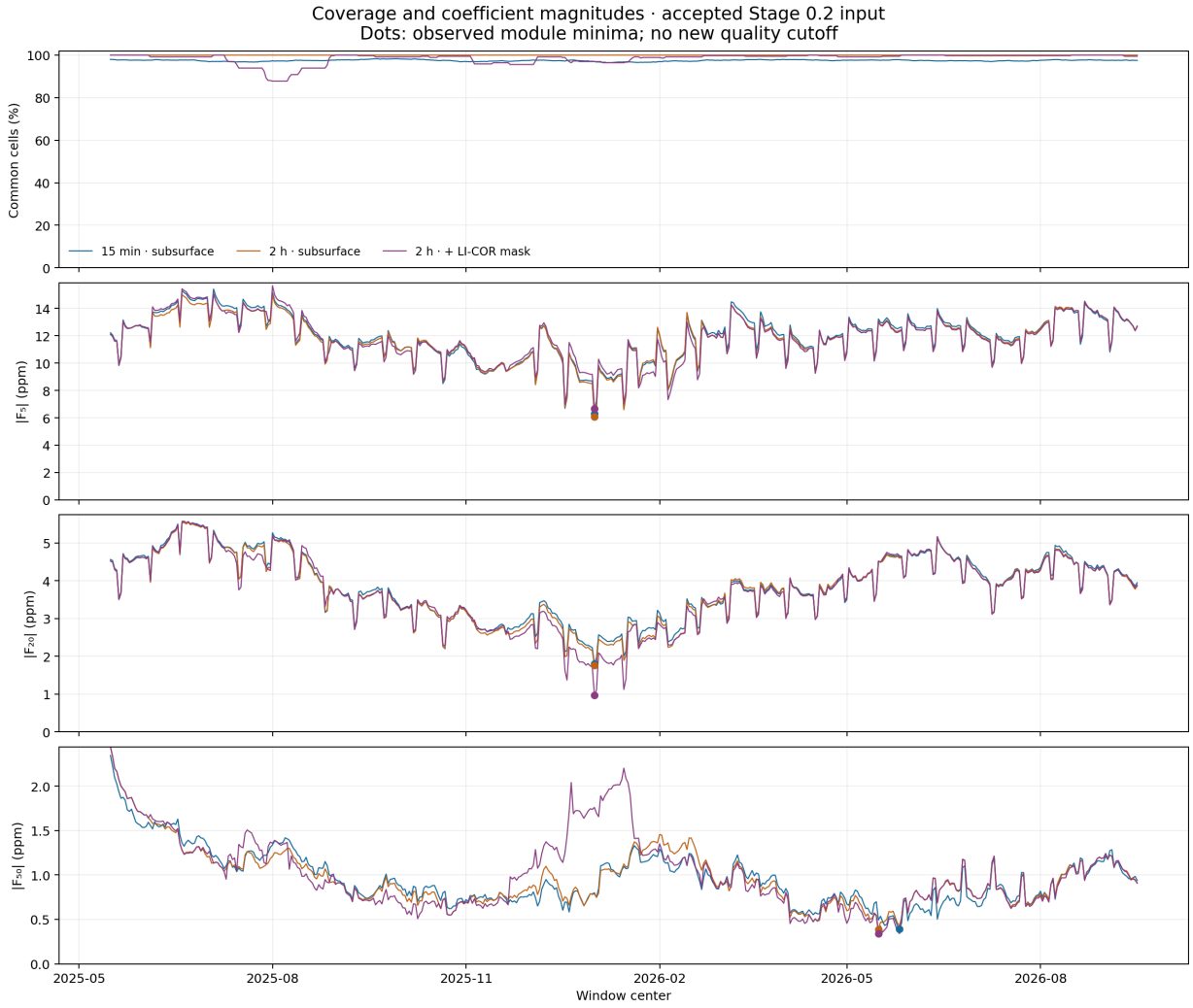


Figure 9: Existing coverage and Fourier-coefficient moduli by grid variant. Low extrema and missingness are shown descriptively; no new admissibility threshold is introduced.

established, extension is

$$\partial_t C = \partial_x(D_{\text{eff}}(\theta, x, t)\partial_x C) + S(x, t), \tag{12}$$

where moisture θ, transport coefficient D_{eff} and combined source/sink term S require their own definitions and measurements. Any physical balance must additionally account for the choice of concentration variable and pore-gas storage. Candidate idealized boundary conditions discussed with I. R. Gabitov are

$$-D_{\text{eff}}\,\partial_x C(0, t) = k[C_{\text{air}}(t) - C(0, t)], \qquad \partial_x C(L, t) = 0. \tag{13}$$

The surface Robin condition is not automatically equivalent to my separate observational assumption $C_0(t) = C_{\text{air}}(t)$. I keep them distinct pending a physical model.

With $\tau = t/T_0$ and $y = x/X_0$ the constant-D reference equation becomes

$$\partial_\tau C = \frac{DT_0}{X_0^2}\partial_y^2 C, \qquad T_0 = \frac{X_0^2}{D_0} \quad (D = D_0 > 0). \tag{14}$$

The illustrative length $X_0 = 0.1$ m does not supply a numerical T_0 without D_0; the scale is not a measured phase delay. None of these models starts future Chapter 6 or implies an estimate of biological activity or chemical interaction [2, 19].

9 Buchstaber framework: approved methodological preparation only

The separate existing Chapter 4 notebook states that application to LEO has not begun. For an admissible uniform scalar series $(f_1, \ldots, f_N)$, the proposed unfolding consists of

$$X_k = (f_k, \ldots, f_{k+n-1})^{\mathsf{T}} \in \mathbb{R}^n, \quad P = N - n + 1, \quad k = 1, \ldots, P. \tag{15}$$

Writing $\bar{X} = P^{-1}\sum_k X_k$ and $S = P^{-1}\sum_k (X_k - \bar{X})(X_k - \bar{X})^{\mathsf{T}}$, with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$, the quadratic affine projection identity is

$$E_q = \min_{\dim L = q} \frac{1}{P}\sum_k \|(I - \Pi_L)(X_k - \bar{X})\|_2^2 = \sum_{i=q+1}^{n} \lambda_i, \qquad L_q^* \in G(n, q). \tag{16}$$

This standard mathematical identity is not a measured LEO result. In particular the Grassmannian $G(n, q)$ is not the phase torus $\mathbb{T}^3$. Synthetic tests, validated data and a separately approved protocol are prerequisites for application to LEO [16, 17, 18].

10 Limitations and open questions

My accepted 0.2 data scope and limited 0.1-B hydrology do not establish a complete operational irrigation-event calendar. The 37 wetting dates provide coarse indirect environmental markers, not exact known forcing times. The earlier 0.3/0.4 illustrations are retained on their old input masks. I have not recomputed the three four-channel phase series for the newly accepted 489-window view. The later 15-minute/two-hour study tests only two subsurface pairs, not a 15-minute atmospheric signal. I have not adopted a cutoff declaring two-hour sampling sufficient, or quantified error against a known true phase. Overlapping windows are dependent and coefficient moduli and real gaps may limit interpretation. The partial Arevalo et al. precipitation dataset does not replace a full operational log. No diffusion coefficient or transport, chemical or biological source has been identified from these observations [2, 9, 13].

11 Next approved steps

I remain at Stage 0.5 before new Chapter 1. The approved immediate discussion with I. R. Gabitov must present the original amplitude spectra, the three preliminary temporal phase series, the 0.5/Q1 example, the accepted 17-month input, hydrological coverage and 37 wetting-day markers, and the new two-pair temporal-resolution diagnostic. The open decision is whether two-hour sampling is sufficient for the selected daily spectral phase question, with explicit consideration of the remaining irrigation-log gaps and the chosen frequency, window, step and phase assessment protocol. I do not automatically launch 15-minute atmospheric reconstruction, machine learning, additional frequencies or physical modeling, or mark Chapter 1 as begun. The independent Buchstaber preparation is not evidence of its application to LEO.

Sources, figures and reproducibility

This report consolidates the source repository at commit `5fb1ca89260192f189d69ac73fa360caec7598d5`, 28.Oct.2026. The existing figures used here, without regeneration, are:

- `output/stage01_irrigation/pilot_hydrology_17months.png`;
- `output/preliminary_fourier_phases/grid_resolution/grid_aggregation.png`;
- `output/preliminary_fourier_phases/grid_resolution/grid_air_support.png`;
- `output/preliminary_fourier_phases/grid_resolution/grid_context.png`;
- `output/preliminary_spectra/four_channel_spectra_17x11.pdf`;
- `output/preliminary_fourier_phases/phase_evolution_17x11.pdf`;
- `output/preliminary_fourier_phases/phase_trajectory_3d.png`;
- `output/preliminary_fourier_phases/jump_diagnostic/phase_jump_local.png`;
- `output/preliminary_fourier_phases/jump_diagnostic/phase_jump_complex.png`.

I preserve the original numerical audit files and their passports rather than generating second versions. In particular, I use the accepted hydrological source `data/stage01_irrigation/pilot_validation.json`, the selected input passport `output/stage02_candidate_selection/passport.json`, and the grid-resolution passport, summary and validation within `output/preliminary_fourier_phases/grid_resolution/`. The original Fourier phase demonstration and Q1 figures retain their earlier masks. Interactive supplementary figures remain in the corresponding `output/` directories. I compile this sole master file from `Project_description/Research_log/` using `latexmk -pdf -interaction=nonstopmode -halt-on-error Research_Report.tex`.

References

[1] CO2Flux, *Project_description/Research_log/README.md*, revision 27.Oct.2026.

[2] CO2Flux, *Project_description/Research_log/PLAN.md*, preparatory Stage 0 and unified report policy, source commit specified above.

[3] CO2Flux, *00_preliminary_co2_spectra.ipynb*, and *output/preliminary_spectra/passport.json*.

[4] CO2Flux, *00_preliminary_fourier_phases.ipynb*, including Section 10 (single Q1 phase jump).

[5] CO2Flux, *output/preliminary_fourier_phases/passport.json*.

[6] CO2Flux, *output/preliminary_fourier_phases/validation.json*.

[7] CO2Flux, *output/preliminary_fourier_phases/jump_diagnostic/phase_jump_validation.json*.

[8] CO2Flux, *details/02_data_checks.md*, verified computational procedures and stated limitations.

[9] CO2Flux, *details/00_irrigation_audit.md*, accepted limited Stage 0.1-B hydrological input, 27.Oct.2026.

[10] CO2Flux, *data/stage01_irrigation/pilot_validation.json*, 14-series hydrological checks.

[11] CO2Flux, *output/stage02_candidate_selection/passport.json*, selected vertical, threshold and 17-month admissibility.

[12] CO2Flux, *output/stage02_candidate_selection/validation.json*, recorded accepted-input checks.

[13] CO2Flux, *output/preliminary_fourier_phases/grid_resolution/grid_passport.json*, three supports, Fourier definitions and limitations.

[14] CO2Flux, *output/preliminary_fourier_phases/grid_resolution/grid_summary.csv* and *grid_validation.json*, 489-window comparisons and tests.

[15] CO2Flux, *01_october_2017.ipynb; 02_monthly_changes.ipynb; 03_long_term_evolution.ipynb*.

[16] CO2Flux, *04_buchstaber.ipynb*, proposed method only.

[17] V. M. Buchstaber, *Time Series Analysis and Grassmannians*, American Mathematical Society Translations, Series 2, vol. 162 (1994), pp. 1–17.

[18] V. M. Buchstaber, *Kolmogorov Widths, Grassmann Manifolds and Unfoldings of Time Series*, Matematicheskii Sbornik, vol. 216, no. 3 (2025), doi: 10.4213/sm10240.

[19] M. Maier and H. Schack-Kirchner, *Using the gradient method to determine soil gas flux: A review*, Agricultural and Forest Meteorology 192–193 (2014), 78–95, doi: 10.1016/j.agrformet.2014.03.006.